CS24309 — THEORY (3.0 CR) MODULE IV NOTES BIT MESRA NEP SCHEME

Module 4: Project Estimation & Scheduling

Albrecht Function Points (FP), Boehm's COCOMO I Equations & PERT / CPM Calculations

Module IV: Software Project Estimation, Metrics & Scheduling

1. Software Project Management Dimensions (People, Product, Process, Project — 4 P's)
2. Size-Oriented Metrics (Lines of Code — LOC) & Inherent Productivity Pitfalls
3. Function-Oriented Metrics: Albrecht's Function Point (FP) Analysis Formulations
4. Unadjusted Function Count (UFC) across 5 Information Domain Characteristics
5. 14 General System Characteristics & Value Adjustment Factor (VAF) Math
6. Boehm's COCOMO I Model: Organic, Semi-Detached & Embedded Project Modes
7. Basic, Intermediate (Cost Drivers / EAF) & Detailed COCOMO Equations
8. Project Scheduling: Work Breakdown Structure (WBS) & Activity Networks
9. Critical Path Method (CPM): Earliest Start/Finish & Latest Start/Finish Slack Times
10. Program Evaluation and Review Technique (PERT) Expected Duration (T_e) & Variance
11. Risk Management Framework: Identification, Projection, Assessment & RMMM Plans
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 3 – 5: Function Point (FP) Analysis

1. Function Point Formula (Albrecht, 1979):

$$\mathrm{FP} = \mathrm{UFC} \times \mathrm{VAF} = \mathrm{UFC} \times \left(0.65 + 0.01 \times \sum_{i=1}^{14} F_i\right)$$

Where UFC is Unadjusted Function Count, and $F_i \in [0, 5]$ are ratings for 14 General System Characteristics (yielding $\mathrm{VAF} \in [0.65, 1.35]$).

Information Domain Element	Low Complexity	Average Complexity	High Complexity
1. External Inputs (EI)	3	4	6
2. External Outputs (EO)	4	5	7
3. External Inquiries (EQ)	3	4	6
4. Internal Logical Files (ILF)	7	10	15
5. External Interface Files (EIF)	5	7	10

Topic 6 & 7: Boehm's COCOMO (Constructive Cost Model)

Basic COCOMO Effort & Schedule Equations:

$$\text{Effort } (E) = a \times (\text{KLOC})^b \quad \text{Person-Months (PM)}$$

$$\text{Development Time } (T_{\text{dev}}) = c \times (E)^d \quad \text{Months}$$

$$\text{Average Staffing (Staff)} = \frac{E}{T_{\text{dev}}} \quad \text{Persons}$$

Project Mode	Characteristics	a	b	c	d
1. Organic	Small teams, familiar in-house domain, relaxed constraints.	2.4	1.05	2.5	0.38
2. Semi-Detached	Medium teams, mixed experience levels, medium constraints.	3.0	1.12	2.5	0.35
3. Embedded	Tight hardware/software constraints, complex safety-critical systems.	3.6	1.20	2.5	0.32

🏛 **Step-by-Step Solved Problem: Basic COCOMO Estimation for a 30 KLOC Semi-Detached Project**

1. **Step 1: Compute Effort (E):**

$$E = 3.0 \times (30)^{1.12} = 3.0 \times 44.91 = \mathbf{134.73} \text{ Person-Months}$$

2. **Step 2: Compute Development Time (T_{dev}):**

$$T_{\text{dev}} = 2.5 \times (134.73)^{0.35} = 2.5 \times 5.54 = \mathbf{13.85} \text{ Months}$$

3. **Step 3: Compute Average Staffing Size:**

$$\text{Staff} = \frac{E}{T_{\text{dev}}} = \frac{134.73}{13.85} \approx \mathbf{9.7} \implies \mathbf{10} \text{ Engineers}$$

Topic 9 & 10: Project Scheduling (CPM & PERT Calculations)

PERT Three-Point Estimation Formulas: - Optimistic time (a), Most likely time (m), Pessimistic time (b).

$$\text{Expected Duration } (T_e) = \frac{a + 4m + b}{6}$$

$$\text{Standard Deviation } (\sigma) = \frac{b - a}{6}, \quad \text{Variance } (\sigma^2) = \left(\frac{b - a}{6}\right)^2$$

🏛️ Top BIT Mesra Exam Questions & Answers (Module IV)

Q1. State the components of an RMMM (Risk Mitigation, Monitoring, and Management) plan. (8 Marks)

1. **Risk Mitigation (Proactive Prevention):** Developing actionable strategy to reduce the probability of risk occurrence.
Example: High developer turnover risk → cross-training team members and establishing thorough documentation standards.
2. **Risk Monitoring (Tracking):** Continuously observing warning indicators to detect when a risk is transitioning from potential threat to reality.
Example: Monitoring team morale, overtime hours, and unresolved bug counts.
3. **Risk Management (Contingency Action):** Executing contingency plans when mitigation fails and the risk actually occurs.
Example: Activating backup external contractors if a key developer departs.